INCH POUND
MIL-DTL-32151 (OS)
14 April 2005

DETAIL SPECIFICATION
IMPULSE CARTRIDGE, CCU-71/A

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for an impulse cartridge to be initiated by one initiated by one shielded mild detonating cord (SMDC) assembly and to actuate the pilot's canopy unlatch thruster which unlocks the canopy before canopy jettison in the F-18 aircraft.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-I-8500	Interchangeability And Replaceability Of Component Parts For Aerospace Vehicles
MIL-P-20444	Primer, Percussion, M42, Parts for and Loading
MIL-D-21625	Design And Evaluation Of Cartridges For Cartridge Actuated Devices

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-464 Electromagnetic Environmental Effects requirements for systems

(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia PA 19111- 5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DOCUMENTS

Naval Surface Warfare Center, Indian Head

WS 21535 Ignition Device, (Percussion), PVU-1/A)

DRAWINGS

Naval Surface Warfare Center, Indian Head

841AS450-1 SMDC Test Tip

850AS826 Closed Bomb Test Fixture

850AS830 CCU-71/A Impulse Cartridge Vibration Test Fixture

(Application for copies should be addressed to Commanding Officer, Naval Surface Warfare Center, Indian Head, Technical Information Branch (Code 4230), Indian Head, MD 20640.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E1742 Standard Practice for Radiographic Examination

(Copies of this document are available online at http://www.astm.org/ or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AS5604 Compass, Magnetic, Pilots Standby

(Copies of this document are available online at http://www.sae.org/, or from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pennsylvania, United States, 15096-001.)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, superseded applicable laws and regulations unless a specific exemption has been obtained.

3. GENERAL REQUIREMENTS

3.1. Preproduction sample. When more than 12 months have elapsed since production of the latest accepted lot, a preproduction sample of 72 cartridges that have met the requirement of 3.2 through 3.5 shall be required. The sample shall be tested as specified in paragraph 4.3.

3.2 Design and construction. The CCU-71/A Impulse Cartridge design and construct shall be identical to that subjected to qualification testing and approved for service use.

3.3 Envelope and installation. The dimensions and configuration of the CCU-71/A Impulse Cartridge shall be as shown on DWG 850AS129. All parts having the same vendor's item number shall be functionally and dimensionally interchangeable in accordance with MIL-I-8500.

3.4 Percussion primer. Primers used in the CCU-71/A Impulse Cartridge shall be of the type originally qualified and from lots manufactured within 24 months of cartridge loading date or from primer lots recertified within 12 months of cartridge loading date.

3.4.1 OEA Part No. 2715500-1. Impulse cartridge shall use two (2) M42CI-PA101 percussion primers manufactured and tested in accordance with MIL-P-20444 (primer, percussion, M42, parts for and loading).

3.4.2 SCOT Part No. 6079900. Impulse cartridge shall use two (2) PVU-1/A ignition devices manufactured and tested in accordance with WS21535 (ignition device, (percussion), PVU-1/A).

3.5 Cartridge inspection.

3.5.1 Visual inspection. All cartridges manufactured under contract shall be visually inspected and screened for defects as outlined in 4.4.

3.5.2 Leakage. Cartridges shall not exhibit a leak rate in excess of 1×10^{-5} cubic centimeters per second of air when tested in accordance with 4.5.

3.5.3 Radiographic inspection. Cartridges shall be free of imperfections in assembly when examined radiographically in accordance with 4.6.

3.5.4 Demagnetization. Cartridges shall not deflect an approved pilot's standby compass more than three degrees in either direction when passed in front of the compass. The cartridges shall be tested in accordance with 4.7.

3.5.5 Electrical bonding resistance. Cartridges shall not exhibit an electrical resistance in excess of 0.0025 ohm when tested in accordance with 4.8.

3.6 Ballistic requirements. Cartridges from each preproduction and production lot shall meet the requirements listed below when subjected to the test prescribed in 4.10.

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3.6.1 Maximum pressure. The upper (u) and lower (l) limits for maximum pressure (psig) when fired over the temperature range from –65 °F to +200 °F shall be U = 2917 psig and L = 1537 psig. See 4.10.1.1.

3.6.2 Ignition delay. The limit for ignition delay shall not be greater than 20 milliseconds. See 4.10.1.2.

3.6.3 Time to maximum pressure. The upper (u) and lower (l) limits for time to maximum pressure (milliseconds) when fired over the temperature range from –65 °F to +200 °F shall be U = 90 milliseconds and L = 40 milliseconds. See 4.10.1.3.

3.6.4 Misfire. There shall be no misfires. See 4.10.1.4.

4. VERIFICATION

4.1 Classification of inspections.

a. Preproduction inspection (see 4.3)
b. Lot acceptance inspection (see 4.4)

4.2 Sampling. Samples for ballistic test shall be selected from the production lot in accordance with Table I. Reduced sample size is recommended after 10 consecutive acceptable lots. Sample plan to be used will be normal unless otherwise specified in the contract. An additional six cartridges shall be provided with each test sample for investigative purposes. Sample cartridges and cartridges made available for investigative purposes shall not be applied as part of the quantity specified for delivery by the contract or order. The production lot size shall be 500 maximum cartridges.

Table I. Conformance sample Size

Lot Size	Normal Sample Size	Reduced Sample Size
2-8	2	2
9-15	3	2
16-25	5	3
26-50	8	5
51-90	13	5
91-150	20	8
151-280	32	13
281-500	50	20

4.3 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 72 cartridges shall be prepared using the same procedures and methods proposed for the production lot. Sixty-six of these cartridges shall be expended in the tests listed herein and six shall be retained for investigative purposes. A preproduction sample acceptable for functional and environmental testing shall be defined as a sample which has met the requirements of paragraphs 3.2 through 3.5 of this specification. The contractor shall deliver the cartridges to the activity designated by the contracting agency for tests. Preproduction sample inspection shall consist of the following tests: leak (72),

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temperature and humidity cycling (18), vibration (12) shock (12), –65 °F ± 5 °F (12), and 200 °F ± 5 °F (12). The numbers in parentheses denote the number of cartridges allotted for each test. All tests are done as prescribed in MIL-D-21625 except vibration which is prescribed in 3.9 herein. The cartridges shall be fired as described in 4.10 and shall meet the functional requirements referenced in 3.6. Any quantity production prior to notification by the contracting agency of preproduction sample acceptability shall be at the contractor's risk. The preproduction cartridges will not be applied as part of the quantity specified for delivery by the contract.

4.4 Cartridges visual inspection. Examine all cartridges and reject those with inadequate or illegible descriptive markings or serial numbers, color coding, dents, deep scratches, or other defects which may prevent proper installation or function.

4.5 Cartridge leakage. Each cartridge in the ballistic sample shall be leak tested in a dry gas leak tester. Cartridges which exhibit a leak rate in excess of 1×10^{-5} cc/sec of air at a pressure differential of one atmosphere ± 0.1 atmosphere shall be considered defective. If one or more cartridges of the ballistic sample leak, then a leak test of each cartridge in the production lot shall be required. If a 100 Percent leak test of the lot is required, it shall be at the expense of the contractor. Cartridges which exhibit a leak rate in excess of that specified above shall be rejected and destroyed.

4.6 Radiographic examination. All cartridges shall be identified with serial numbers prior to radiographic examination. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The cartridges shall be positioned to give the most revealing cross-section of the cartridge. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the radiographic plate. Defective cartridges found by the vendor shall be clearly identified on the radiographic plate and those defective shall be removed from the production lot. Radiographs or copies of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. (see 5.1).

4.6.1 X-ray examination. Each cartridge shall be x-rayed in accordance with ASTM E1742, to confirm the presence and proper assembly of all parts. Any observable imperfections in assembly shall be cause for cartridge rejection.

4.6.2 N-ray examination. Each cartridge shall be n-rayed to confirm the presence of the propellant, non-metal parts, no foreign hydrogenous materials and the proper assembly of pyrotechnic/non-metallic materials. Any observable imperfection shall be cause for rejection.

4.7 Demagnetization. Each cartridge shall be moved past a pilot’s standby compass face (in accordance with SAE AS5604) by placing the cartridge six inches from the compass face, in a north-south horizontal position with the compass set in an east-west heading. The cartridge and the compass shall be placed at the same height from the floor. The cartridges shall be moved past the compass face in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved back past the compass face. This procedure shall be repeated for each 90 degrees of the cartridge.

4.8 Electrical bonding resistance. Each cartridge shall be tested using a four point probe milliohmmeter in accordance with MIL-STD-464. The electrical bonding resistance shall be measured between the thread of the SMDC inlet port and the thread at the outlet end of the cartridge.

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4.9 Vibration. The vibration test shall be performed on all 12 cartridges at the same time. The twelve cartridges shall be mounted in the vibration test fixture NAVAIR Drawing No. 850AS830 for the CCU-71/A Impulse Cartridge. The test specimens shall be exposed to the sinusoidal vibration environment shown in Figure 1 and Table II with the frequency varying logarithmically from 5 Hz to 2,000 Hz and returning to 5 Hz in approximately 20 minutes. Each specimen, will be exposed to the environment for one hour in each of the three mutually perpendicular axes, at ambient, +200°F, and –65°F, for a total of nine hours of testing.

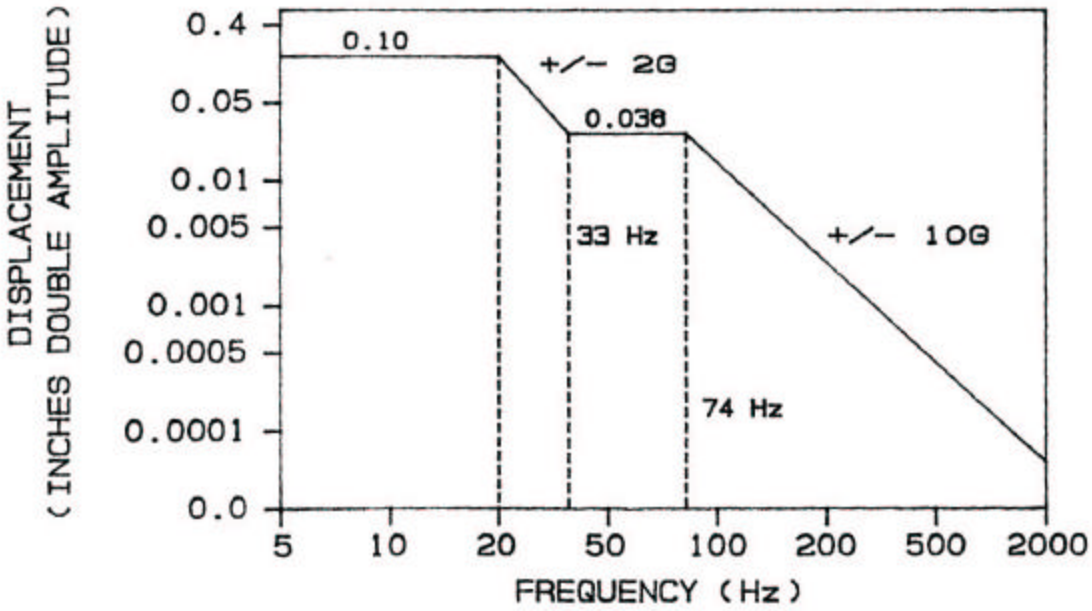


Figure 1. Non Gunfire Vibration

Table II. Sinusoidal vibration environment

Frequency (Hz)	Acceleration/Amplitude
5 to 20	0.10 inch D.A.
20 to 33	±2 g's peak
33 to 74	0.036 inch D.A.
74 to 2,000	±10 g's peak

4.10 Ballistic test. Cartridges shall be selected from each production lot in accordance with paragraph 4.2. The sample of cartridges shall be divided, as nearly as possible, into three equal groups. If the sample of cartridges is not equally divisible by three, the excess cartridge(s) shall be placed in the group that is to be fired at –65 °F ± 5 °F. One group each shall be conditioned for a period of time neither less than 4 hours nor greater than 24 hours at –65 °F ± 5 °F and +200 °F ± 5 °F. The other group shall be conditioned for a period of time not less than 4 hours at +70 °F ± 5 °F. The cartridges shall be fired in the closed bomb test fixture NAVAIR drawing no. 850AS826, within 3 minutes after removal from temperature conditioning. The cartridges shall be temperature conditioned with the SMDC test tip (NAVAIR drawing no. 841AS450-1) installed. If any cartridge is exposed to the temperature environment for greater than the time allowed or not fired within 3 minutes after removal, it shall be returned to a +70 °F ± 5 °F

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environment for at least 4 hours and then reexposed to the original environment for not less than 4 hours and not more than 24 hours before test firing. The reexposure may be performed as many times as necessary to complete the required ballistic test. Instrumentation shall be provided to measure and record the cartridge pressure (psig) in the test chamber; ignition delay (milliseconds); and time to maximum pressure (milliseconds). The test chamber shall be thoroughly cleaned after each cartridge firing. The production lot shall be accepted if the sample fulfills the following criteria.

4.10.1 Acceptance criteria.

4.10.1.1 Maximum pressure. The maximum pressure shall be measured at the point at which the pressure reaches its maximum in the test fixture. The lot shall be acceptable with respect to maximum pressure if the ballistic sample meets the requirements of 3.6.1.

4.10.1.2 Ignition delay. The ignition delay shall be measured from the actuation of the blasting cap to first indication of pressure. The lot shall be acceptable with respect to ignition delay if the ballistic sample meets the requirements of 3.6.2.

4.10.1.3 Time to maximum pressure. The time to maximum pressure shall be measured from the first indication of pressure to the point at which the pressure reaches its maximum in the test fixture. The lot shall be acceptable with respect to time to maximum pressure if the ballistic sample meets the requirements of 3.6.3.

4.10.1.4 Misfire. A cartridge that does not fire under the conditions specified in 4.10 shall be considered defective, and the lot considered unacceptable, unless the misfire is clearly attributable to faulty test equipment or procedure.

4.10.2 Retest. There shall be no retests.

4.10.3 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridges, the original test results shall be discarded and that test reconducted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

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6.1 Intended use. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems. The assembly was designed for use in military aircraft and has no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

A. Title, number and date of the source control drawing and this specification
B. Whether preproduction inspection is required, and if so, specify the test activity
C. Production lot size, sample size and test activity
D. Direction for shipping radiographic plates of entire production lot along with ballistic sample (paragraph 4.6) to the activity conducting production lot acceptance test
E. Items of data required for each preproduction and production lot

6.3 Safety precautions. The safety precautions requirements of the "contractor's safety manual for ammunition, explosives and related dangerous material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the defence acquisitions regulation (DAR) 1-323. When this specification is used as part of the description of work to be accomplished by a government activity, the safety precaution requirements of "ammunition and explosives ashore" (OP 5) should be made applicable.

6.4 Explanation of terms. For the purpose of this drawing, NAVSUPINST 4030.28A (20 June 1975) paragraphs 1-4, 2-2, 2-3, and appendix A, when applicable should be used to clarify terms.

6.5 Subject term (key word) listing.

Ejection systems, aircrew
Canopy jettison
SMDC
F-18

Custodian:
Navy-OS

Preparing activity
Navy - OS
(Project 1377-0239)

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